

Johan Benedict A. Cristobal

JohanBenedict.Cristobal[at]lmu.edu

www.johanmath.com

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EDUCATION

- 2025 **Ph.D. in Mathematics** at University of Nebraska - Lincoln (UNL)
Area/Focus: Mathematics Education
Dissertation: *Culture and Context: How Frames of Teaching and Learning Mathematics Form and Change for Graduate Student Instructors*; Advised by Dr. Yvonne Lai
- 2021 **M.S. in Mathematics** at UNL
- 2020 **B.S. in Mathematics of Computation** at University of California, Los Angeles

EMPLOYMENT

- 08/2025 - **Visiting Assistant Professor**
Dept. of Mathematics, Statistics and Data Science, Loyola Marymount University
- 05/2025 - 08/2025 Postdoctoral Researcher
Center for Science, Mathematics and Computer Education (CSMCE) at UNL
- 05/2021 - 05/2025 Graduate Research Assistant — CSMCE at UNL
- 08/2020 - 05/2025 Graduate Teaching Assistant — Dept. of Mathematics, UNL

TEACHING

Courses Taught at Loyola Marymount	Term (Sections)
• Math 112 - Calculus for Business	Spring 2026 (3), Fall 2026 (1)
• Math 120 - Precalculus Mathematics	Fall 2025 (2), Spring 2026 (1), Fall 2026 (2)
• Math 101 - Algebra	Fall 2025 (2), Fall 2026 (1)

at Nebraska - Lincoln | ♦ denotes courses I also worked on as Course Developer

- Math 221 - Differential Equations (Tutor) Summer 2022, Summer 2025
- ♦ Math 103 - College Algebra and Trigonometry Fall 2024
- ♦ Math 97 - Assisting Learning for University Mathematics Fall 2024
- Math 302 - Math Modeling Spring 2023
- ♦ Math 101C - College Algebra Corequisite Fall 2022
- ♦ Math 101 - College Algebra Spring 2022
- ♦ Math 100A - Intermediate Algebra Fall 2021
- Math 804T - Experimentation, Conjecture and Reasoning (Assistant Teacher) Summer 2021
Virtual Synchronous Course for Middle School Teachers in an MAT program

Walter Mientka Teaching Award | Awarded Fall 2021 by the UNL Department of Mathematics

“The Walter Mientka Award is given to students who demonstrate exceptional promise as a teacher based on their work during their first one or two years as a graduate teaching assistant in our department.”

Qualifying Exam Workshop Review and Preparation Sessions (UNL) Summer 2022

Math 830/831 - Ordinary and Partial Differential Equations Workshop Leader

Note: Also was a Grader for Math 830 (Fall 2023) and Math 831 (Spring 2025).

PAPERS

Peer-reviewed Journal Articles

- [1] Black, K., Smith, W. M., Hagman, J. E., Tremaine, R., **Cristobal, J. B.**, & Fantin-Hardesty, K. (under review). Tugging on Threads: A Loving Self-Critique of our White Comforts.
- [2] **Cristobal, J. B.** (under review). Cultures of exclusion in mathematics education pervades from a property view of mathematics knowledge.
- [3] **Cristobal, J. B.**, Lai, Y., Young, C., Bonnesen, C., Strayer, J. F., Anhalt, C., Casey, S., Kohler, B., Lischka, A. E., & Ross, A. (under review). Instructional Activities in Tertiary Mathematics Courses: What (and From Which Component) Do Prospective Secondary Teachers Report as Beneficial.
- [4] **Cristobal, J. B.**, Pai, L., Black, K., McNeill, R. T., Akin, V., & Hagman, J. E. (under review). Leveraging Ambiguous Definitions of Rigor to Promote Equity in Instruction
- [5] Funk, R., Pai, L., **Cristobal, J. B.**, & Xie, X. (under review). “I’ve Had to Show a Lot of Tenacity”: Stories of Power and Empowerment in the Experiences and Identities of S-STEM Women Scholars.
- [6] McNeill, R. T., Hagman, J. E., Black, K., Kress, N., Tremaine, R., Pai, L., & **Cristobal, J. B.** (under review). Theorizing Educational Microcosms of Power: Interrogating Rigor and its Instructional Mechanisms of Inequality in Undergraduate Mathematics.

Peer-reviewed Reports in Conference Proceedings

- [7] **Cristobal, J. B.** (2024). Coloring the Relationship of Frames and Responses in Teacher Noticing. In Cook S., Katz B., Moore-Russo, D. (Eds.), *Proceedings of the 26th Annual Conference on Research in Undergraduate Mathematics Education* (pp. 268-276). Omaha, Nebraska. [Link to article](#).
- [8] Funk, R., Pai, L., & **Cristobal, J. B.** (2024). “Persistence in a S-STEM grant: Understanding the Intersectional Experiences of Women Pursuing STEM.” Conference Paper for the *2024 American Society for Engineering Education Annual Conference & Exposition*.
- [9] Bennett, A. B., **Cristobal, J. B.**, Lai, Y., & Young, C. (2025). Real Analysis Professors’ Noticing of Equity Gaps and Minoritized Students’ Perspectives. In S. Cook, B. P. Katz & K. Melhuish (Eds.), *Proceedings of the 27th Annual Conference on Research in Undergraduate Mathematics Education* (pp. 624-633). Alexandria, VA. [Link to article](#).
- [10] **Cristobal, J. B.** (2025). From “Struggle” to “Acceptance”: Andy’s Narrative of First-Time Teaching and Her Frames. In S. Cook, B. P. Katz & K. Melhuish (Eds.), *Proceedings of the 27th Annual Conference on Research in Undergraduate Mathematics Education* (pp. 149-156). Alexandria, VA. [Link to article](#).
- [11] **Cristobal, J. B.** (2025). Aspects of Culture and Context which Shape Frames of Teaching and Learning. In S. Cook, B. P. Katz & K. Melhuish (Eds.), *Proceedings of the 27th Annual Conference on Research in Undergraduate Mathematics Education* (pp. 157-164). Alexandria, VA. [Link to article](#).
- [12] Black, K., **Cristobal, J. B.**, Hagman, J. E., Fantin-Hardesty, K., Smith, W. M., & Tremaine, R. (2026). Tugging on threads: a loving self-critique of our (white) comforts. In Adiredja, A. P., Katz, B. P., Melhuish, K., & Gallagher, L. (Eds.), *Proceedings of the 28th Annual Conference on Research in Undergraduate Mathematics Education* (pp. 54-62). Alexandria, VA. [Link to article](#).
- [13] **Cristobal, J. B.**, Tremaine, R., Black, K., Hagman, J. E., Kress, N., McNeill, R. T., & Pai, L. (2026). When we say “rigor” in undergraduate mathematics, what do we mean?. In Adiredja, A. P., Katz, B. P., Melhuish, K., & Gallagher, L. (Eds.), *Proceedings of the 28th Annual Conference on Research in Undergraduate Mathematics Education* (pp. 1018-1023). Alexandria, VA. [Link to article](#).
- [14] Hagman, J. E., McNeill, R. T., Tremaine, R., Pai, L., Kress, N., Black, K., & **Cristobal, J. B.** (2026). Power implicit in discourses of mathematical rigor within an equity-oriented improvement community. In Adiredja, A. P., Katz, B. P., Melhuish, K., & Gallagher, L. (Eds.), *Proceedings of the 28th Annual Conference on Research in Undergraduate Mathematics Education* (pp. 144-152). Alexandria, VA. [Link to article](#).

SERVICE

Research in Graduate Mathematics Education (RGME)

Website Manager, Member on the Organizing Committee
Writer for the 2025 Working Group Proposal

Summer 2024 - Present
<https://rgme-group.com/>

UNL Math Department Recruitment — Graduate Student Volunteer (various dates)

- Math Alliance Field of Dreams Conference (November 2022)
- Joint Mathematics Meeting (January 2024)
- Nebraska Conference for Undergraduate Women in Mathematics (2024, 2025)

UNL Math Department Orientations

Presenter in the First Year and the Returning Graduate Student Orientation,
Organizer of the Mathematics Learning Assistant Orientation

Summer 2024

UNL Math Day Organizing Committee

Organizer for Volunteer outreach and scheduling

Fall 2023 - Spring 2025

Journals Refereed/Reviewed For

- *PRIMUS: Problems, Resources, and Issues in Mathematics Undergraduate Studies*
- *TEEM: Teaching for Excellence & Equity in Mathematics*

GRANT-FUNDED WORK

- **Achieving Critical Transformations in Undergraduate Programs in Mathematics (ACT UP Math) NSF ECR-2201486** (terminated by DOGE on April 2025)

January 2025 to Present – A multi-institution research project to identify the mechanisms and structures that best support departments and instructors in making data-informed decisions to promote DEI and critically transform introductory mathematics courses and programs. **My contribution:** Supported research analysis, writing, and propagation efforts

- **Educating Undergraduate Students for STEM Career Opportunities in Nebraska: Networks, Experiential-learning, & Computational Thinking (STEM CONNECT) NSF/S-STEM DUE-1930211**

Spring 2024 to Fall 2025 – an S-STEM scholarship program that provided curricular and co-curricular support to academically talented, low-income students with interest in careers that require strong skills in mathematics. **My contribution:** Supported research writing and propagation efforts

- **First Generation First Year Research Experience for Undergraduates (REU) NSF DMS-2236983**

Summer 2024 – A summer REU for a group of four incoming first-generation freshman from an Upward Bound program. **My contribution:** Co-organized and co-led with another graduate student (Michael Pieper). Building on recent work by Mikil Foss and Michael Pieper (2025; in progress) which showed a classically ill-posed problem in calculus of variations is well-posed if interpreted using nonlocal operators, the REU project tackled the questions: *Can this result be used in practice in the context of signal processing? Under what kinds of conditions would it yield a computational result?* At the forefront of the REU, we provided orientation-like experiences to students with touring the UNL campus and hosting panels.

- *Understanding the experiences of underrepresented students in 300-level mathematics courses*
College of Arts & Sciences Strategies Priorities (UNL Internal Grant)

Fall 2023 to Spring 2025 – Using interview data with students to understand how equity changes can take place inside the program. **My contribution:** Supported data analysis efforts for the purpose of creating and running a follow-up study with mathematics professors and how they notice and frame these narratives with respect to their practice.

- Mathematics of Doing, Understanding, Learning and Educating for Secondary Schools (MODULE(S²)) – NSF DUE-1726723, 1726098, 1726252, 1726744, and 1726804

Summer 2022 to Spring 2024 – A program created, supported, and examined the use of undergraduate mathematics course materials that provide opportunities for users to develop deep mathematical knowledge as it relates to doing the work of teaching. **My contribution:** Supported analysis and writing efforts to characterize increased confidence to succeed in teaching, doing, or learning mathematics, or value for inclusive teaching practices.

- Investigating the Role of Collaboration on the Development of Student Ideas using a Learning Progression for the Function Concept – NSF DRL-2101393

Spring 2023 to Spring 2024 – a research project that examined how students collaborated on problems using an online platform. **My contribution:** Supported study design and data collection efforts

- CSForAll: Adapt, Implement and Research at Nebraska – NSF DRL-1837476

Summer 2021 – a research project that examined the adaptation and implementation of a validated K-8 Computer Science curriculum. **My contribution:** Supported investigation (data collection and curation) efforts and gave feedback on the K-12 computer science curriculum being developed.

GRANTS & AWARDS

- [1] *Simons Laufer Mathematical Sciences Institute's 2024 Travel Grant for Critical Issues in Mathematics Ed.*
- [2] *American Mathematical Society's 2025 Joint Mathematics Meetings Graduate Student Travel Grant*
- [3] *University of Nebraska-Lincoln's Graduate Student Travel Award for Spring 2025*
- [4] *Loyola Marymount University's Spring 2026 NTTF Professional Development (Delphi) Award*

SELECT PRESENTATIONS

- [1] *One graduate students' experience teaching for the first time Students Teaching*
Joint Mathematics Meeting 2025 in the AMS Contributed Papers, Mathematics education.
<https://youtu.be/gFIJFILrZkE>
- [2] *Success, Struggle, Surprise, and Short-term Goals: Reflections from Mathematics Graduate Students Teaching*
Joint Mathematics Meeting 2025 in the AMS Special Session: Research Presentations by Math Alliance Scholar Doctorates. https://youtu.be/ht0_geTrGRc
- [3] *Successes, Struggles, Surprises and Short-term Goals of Mathematics Graduate Student Instructors Teaching for the First Time*
Mathematical Association of America's MathFest 2024 in the Research in Undergraduate Mathematics Education session. <https://youtu.be/1TYHb2xk5Hg>
- [4] *Understanding the Intersectional Experiences and Identities of Women Who Persist in STEM*
2024 Nebraska Mathematical Association of Two Year Colleges, with Dr. Rachel Funk.

SELECT WORKSHOPS & PROFESSIONAL DEVELOPMENT ATTENDED

\$ = Travel funded

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| 1. paraDIGMS 2021 Fall Conference Virtual | October 2021 |
| 2. Field of Dreams Conference Virtual | November 2021 |
| 3. 6th Northeastern Conference
on Research in Undergraduate Mathematics Education (RUME) Virtual | October 2022 |

4. \$ Math Alliance Field of Dreams Conference Minneapolis, MN	November 2022
5. 25th Annual Conference on RUME Omaha, NE	February 2023
6. 26th Annual Conference on RUME Omaha, NE	February 2024
7. \$ Critical Issues in Mathematics Education 2024: Bringing Innovation to Scale: Teaching-Focused Faculty as Change Agents Berkeley, CA	April 2024
8. \$ Mathematical Association of America's MathFest Indianapolis, IN	August 2024
9. \$ Joint Mathematics Meetings Seattle, WA	January 2025
10. \$ 27th Annual Conference on RUME Alexandria, VA	February 2025
11. Midwest Numerical Analysis Day 2025 Lincoln, NE	April 2025
12. Workshop for Research and Workforce Development in Fluid Mechanics Lincoln, NE	May 2025
13. \$ 28th Annual Conference on RUME Alexandria, VA	February-March 2026

NOTABLE COURSE WORK

■ UNL's Department of Mathematics:

833 - Nonlinear Optimization

934 - Topics in Partial Differential Equations (Fluid Dynamics)

941 - Partial Differential Equations (Distributions, Sobolev Spaces, and Elliptic BVPs)

958 - Topics in Combinatorics (Geometry and Probability in High Dimensions for Data Science)

■ UNL's Department of Educational Psychology:

859 - Statistical Methods

Quantitative Research Methods

900K - Qualitative Approaches to Education Research

Qualitative Research Methods

■ UNL's Department of Teaching, Learning, and Teacher Education:

801 - Curriculum Inquiry

807C - Equitable Practices in Mathematics Education: Mathematics Classroom Discourse

861 - Education for a Pluralistic Society: Foundation and Issues

880E - Teaching with Technology: Instructional Technology in Mathematics

949B - Critical, Anti-colonial, & Decolonizing Theories in Education

Qualitative Research Methods

SKILLS

Mathematics $\text{\LaTeX}$, curriculum development

Statistical Methods RStudio

Programming C, C++, Python, Matlab, LISP, SQL, and HTML/CSS (Web-design)

Languages English (fluent), Filipino/Tagalog (native fluent), Spanish (intermediate)

Microsoft Suite Word, Excel, Powerpoint

Adobe Creative Suite Acrobat, Photoshop, Illustrator, Premiere Pro, InDesign, Lightroom

Video/Content Creation <https://www.youtube.com/@johango>, textbook/open educational resource videos, and conference or presentation videos